

RUSSELL DUDEK

ENTERPRISE AI WORKFLOW ENGINEERING | AGENTIC PROCESS AUTOMATION | GOVERNED AI ENABLEMENT

Enterprise Engineer Sr - AI | Requisition 71280

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Operator-technologist who makes complex work visible, connects process owners and engineering teams, and builds governed, testable AI-assisted workflows. Direct experience eliciting requirements, mapping current and future states, shaping prompt and agent workflows, defining evaluation and human-review controls, and measuring operating impact in technical and non-technical terms.

CORE MATCH

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|--------------------------------|---|--|---|
| • AI prompt engineering | • Generative AI tools and enterprise copilots | • Prompt libraries, templates and scaling | • Agentic coding solutions and AI agents |
| • AI-assisted productivity | • Automation and workflow enhancement | • Business process mapping | • Requirements elicitation |
| • Process reimagination | • Quality engineering and evaluation | • Security-testing awareness and risk controls | • SOLID engineering principles |
| • Human-in-the-loop governance | • Python, SQL and JavaScript | • RAG and structured context | • Technical / non-technical communication |

DIRECTLY RELEVANT EXPERIENCE

Vape-Jet | Director of Operations 2025-Present

- Build and refine AI-assisted workflows across production, purchasing, quality, customer success, support, engineering issue flow, ERP/Odoo, Jira, HubSpot, Slack, telemetry, and cross-functional operating processes.
- Elicit requirements from process owners, map workflow friction and handoffs, and translate needs into dashboards, standard work, training, automation opportunities, escalation paths, and measurable operating mechanisms.
- Connect factory, field, customer, and machine-fleet signals from a software-enabled robotics environment with approximately 1,000 fielded robots to earlier risk detection, prioritized work, cleaner handoffs, and continuous improvement.
- Apply quality engineering, root-cause, validation, human review, and governance practices so AI-assisted productivity improves work without obscuring accountability.

DudeWorth | Founder / AI and Automation Advisor 2017-Present

- Design AI-readiness and automation frameworks that assess business value, feasibility, data readiness, governance, adoption burden, reuse, time-to-value, and organizational impact before work is automated.
- Create agentic workflow patterns using structured context, retrieval / RAG, AI agents, prompts, human judgment, escalation, decision records, evaluation criteria, and governance rails.
- Develop concepts and prototypes for agentic operating models that help process-intensive groups reimagine work, streamline decisions, and convert repeatable tasks into reusable AI-assisted workflows.
- Advises on AI-enabled operations, logistics technology, warehouse automation, autonomous mobile robots, transportation systems, network analysis, and workflow design.

Amazon | Operations Management / Site Leader 2014-2018

- Worked hands-on in a 24/7 customer-backward operation supporting approximately five million second-day orders annually, using Lean and Gemba to improve safety, quality, throughput, and customer experience.
- Built process visibility, standard work, labor-planning mechanisms, escalation paths, and operating routines that made high-volume work easier to understand, measure, and improve.
- Contributed to logistics innovation through Prime Air research and Amazon Flex geolocation concepts, translating operating problems into technology-enabled process improvements.

Compunetics | Director of Operations and Engineering Management 2000-2013

- Progressed through R&D program management, production management, strategic acquisition, operations management, and engineering-management responsibilities in advanced electronics and high-reliability environments.
- Worked across engineering, manufacturing, quality, customer delivery, and complex technical programs; built quality systems, standard work, root-cause practices, performance measures, process controls, and disciplined change mechanisms.
- Translated complex technical information for varied audiences and collaborated with high-consequence customers and partners including DoD, CERN, Intel, and AMD.
- Published industry articles, delivered technical talks, and participated in the HyperTransport Consortium.

COMPLEMENTARY ENGINEERING & TRANSFORMATION EXPERIENCE

ZeusVu | Chief Pilot, Autonomous Systems and Aerial Data Operations 2018-2022

- Led regulated drone operations across medical, energy, logistics, real-estate, and restricted-airspace contexts with a 100% safety record and no FAA incidents.
- Developed TensorFlow-based concepts for automated aerial inventory and computer-vision-enabled field intelligence.
- Designed medical drone delivery and chain-of-custody concepts connecting technology, risk, stakeholders, security-conscious controls, and governed execution.

Cardinal Building Products | Regional Director 2021-2023

- Built the company's first remote regional operation and modernized distributed customer, CRM, commercial, fulfillment, logistics, digital marketing, SEO, and online-buying workflows.
- Advanced AI-enhanced transportation-management and route-planning concepts by connecting business requirements, process mapping, data, and user adoption.

Enviro Social Capital | Chairman of the Board 2014-2017

- Co-founded a public-private impact initiative focused on sustainable infrastructure, green finance, and civic transformation; helped shape and evaluate a \$200 million Social Impact Bond feasibility concept.

IEEE | CMPT and EDS Pittsburgh Chapter Chair 2012-2016

- Organized technical programming, conferences, seminars, and industry-academic engagement across electronics, manufacturing, STEM outreach, and emerging technology.

SELECTED AI ENGINEERING MECHANISMS

Prompt system design Translate business intent into reusable system, role, context, constraint, example, output-schema, and evaluation components; iterate against representative cases rather than one-off demonstrations.	Agentic process architecture Separate responsibilities, tools, state, retrieval, human authority, exception handling, and escalation so agentic coding solutions remain modular, understandable, and aligned with SOLID principles.
Quality and security testing Test expected outputs, ambiguity, edge cases, prompt-injection exposure, data boundaries, failure modes, abstention, traceability, and human-review triggers before scaling.	Automation value model Baseline the current process and quantify cycle time, touch time, rework, exception rate, quality, decision latency, adoption, retained use, and organizational impact after change.

HANDS-ON AI ENGINEERING APPROACH

1. Elicit & Map Interview process owners, clarify requirements, identify decision points, exceptions, current-state friction, and measurable baselines.	2. Prompt & Prototype Write, test, and iterate prompts; structure context; select agent, retrieval, automation, or assistive patterns; keep scope modular and SOLID.	3. Evaluate & Secure Define expected outputs, quality checks, failure modes, security-testing considerations, human authority, escalation, and evidence needed for trust.
4. Embed & Educate Integrate the smallest useful workflow, document standard work, demonstrate the solution, and communicate in technical and non-technical terms.	5. Measure Value Quantify cycle time, touch time, rework, exception rate, quality, decision latency, adoption, retained use, and organizational impact.	6. Scale Responsibly Convert validated prompts and patterns into reusable assets, monitor performance, improve through feedback, and stop what does not create value.

EDUCATION, CREDENTIALS & TECHNICAL TOOLKIT

EDUCATION University of Pittsburgh Bachelor of Science	CREDENTIALS • Google AI Essentials Certification • IBM Enterprise Design Thinking Practitioner • Six Sigma Certification • OSHA 10	TECHNICAL TOOLKIT Agentic AI LLM workflows Prompt engineering RAG Human-in-the-loop design Python SQL JavaScript TensorFlow Tableau ERP/Odoo Jira Confluence HubSpot Slack GIS	ROLE-RELEVANT SCALE ~1,000 robots Machine-fleet feedback loops. ~5M orders / year Scaled 24/7 operations. 100% flight safety No FAA incidents. Cross-functional AI work Operations, engineering, support, quality, and workflow design.
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Individual-contributor fit: The through-line is direct mechanism-building rather than hierarchy: observe the work, elicit requirements, map the process, design and test the prompt, retrieval, automation, or agentic workflow, define controls, demonstrate the result, measure changed work, and iterate with users and engineers.